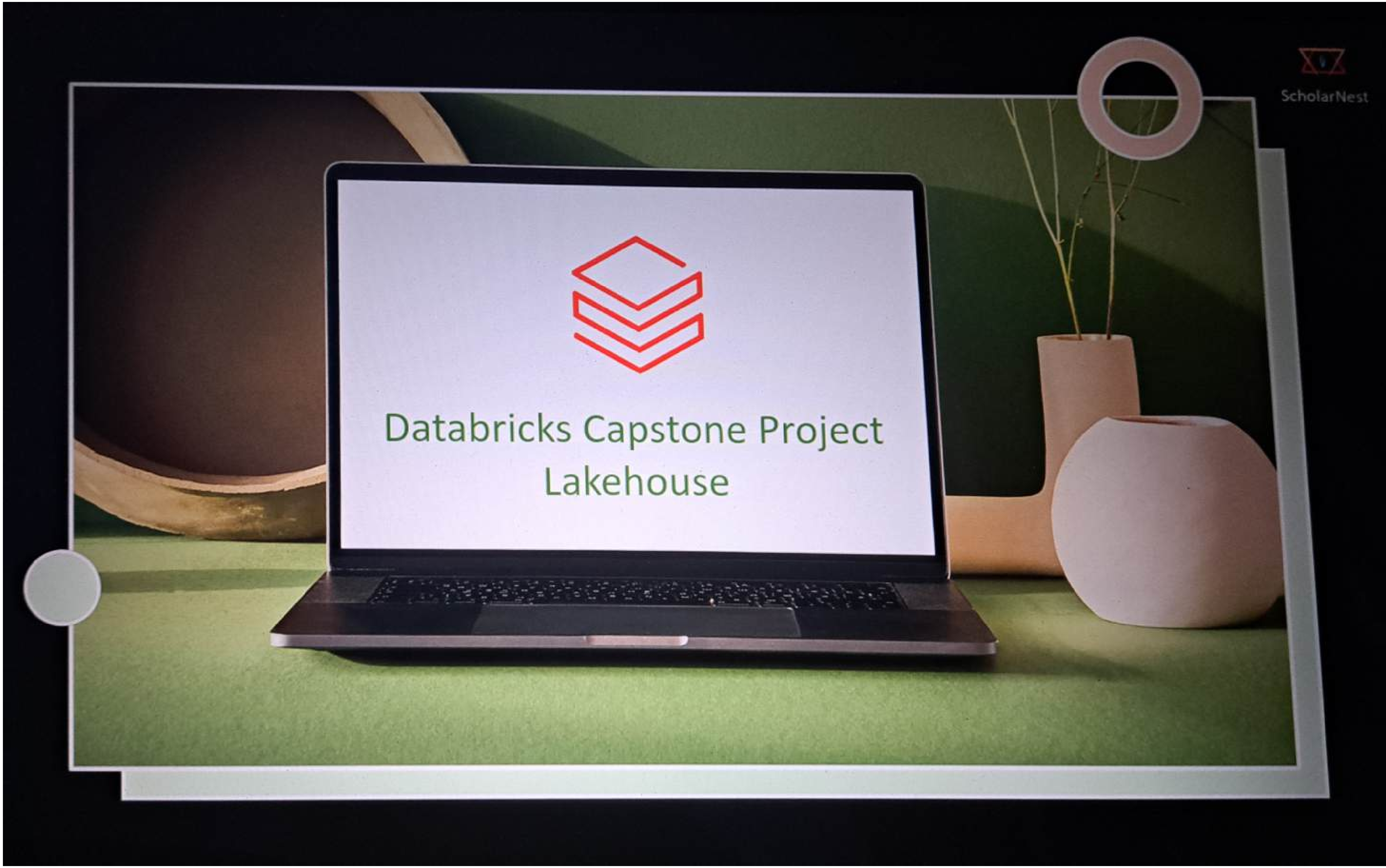
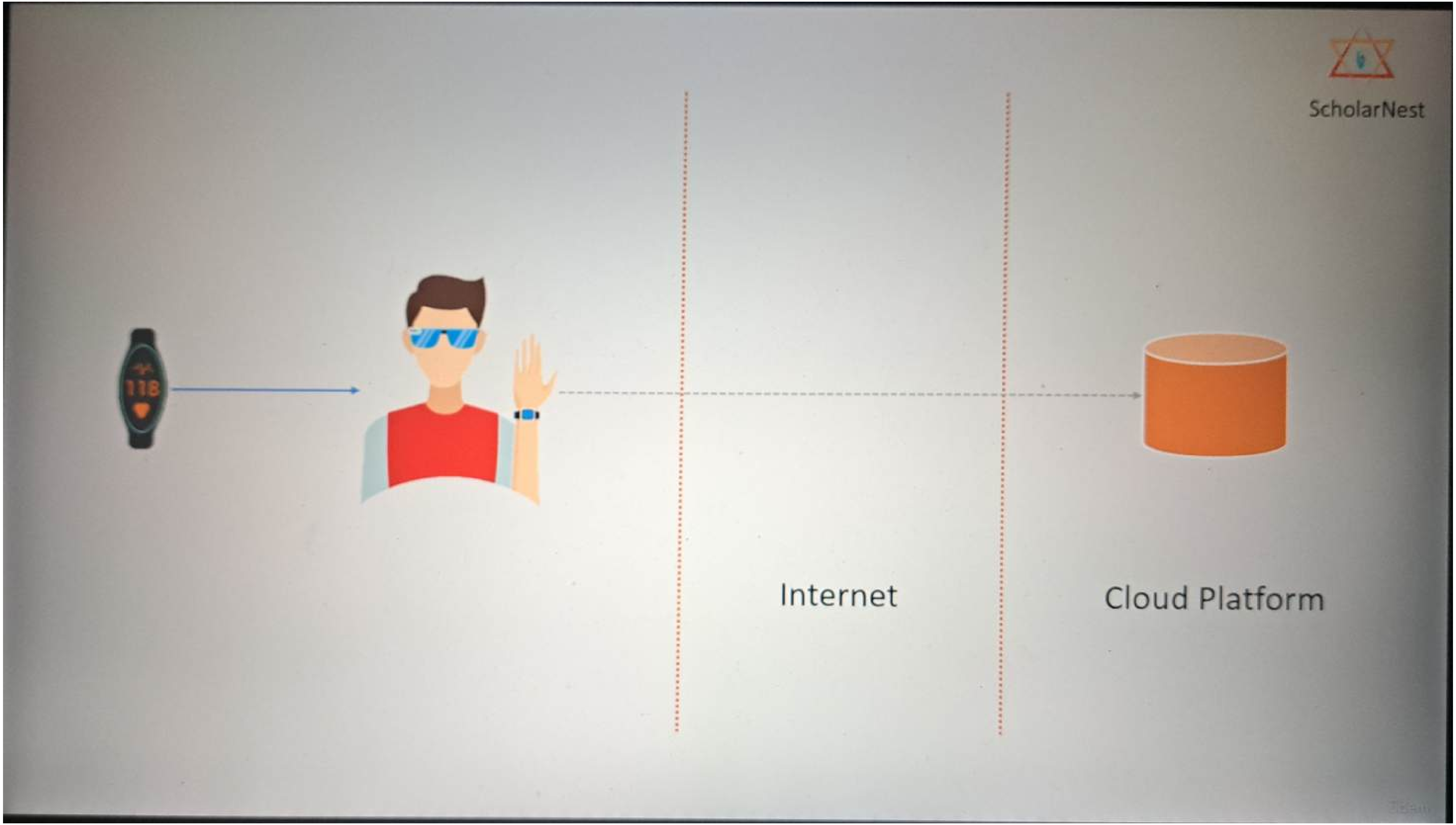
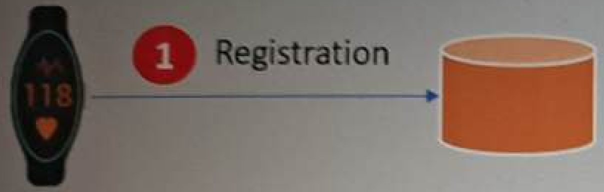

ScholarNest
Databricks Capstone Project
Lakehouse

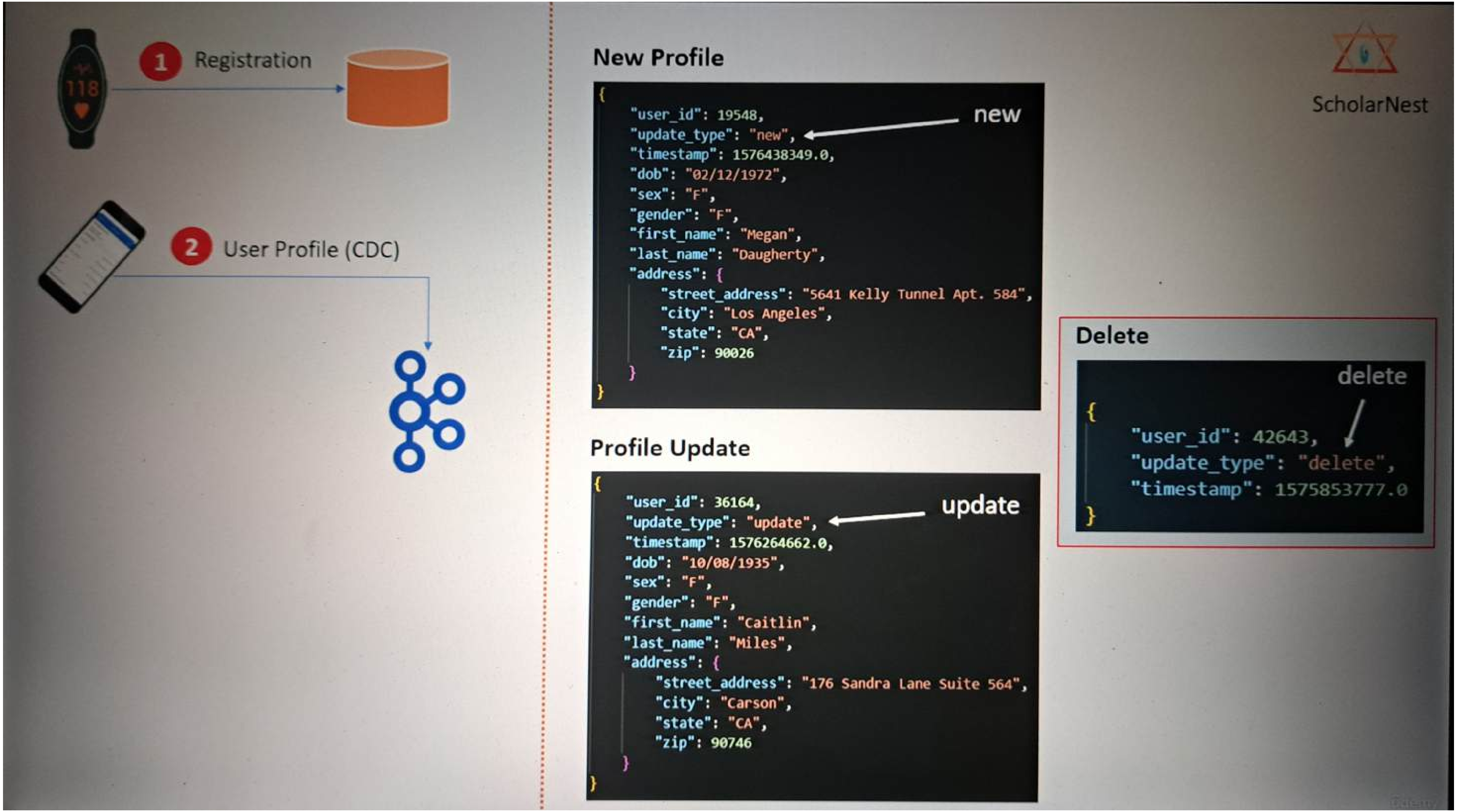

ScholarNest
118
Internet
Cloud Platform

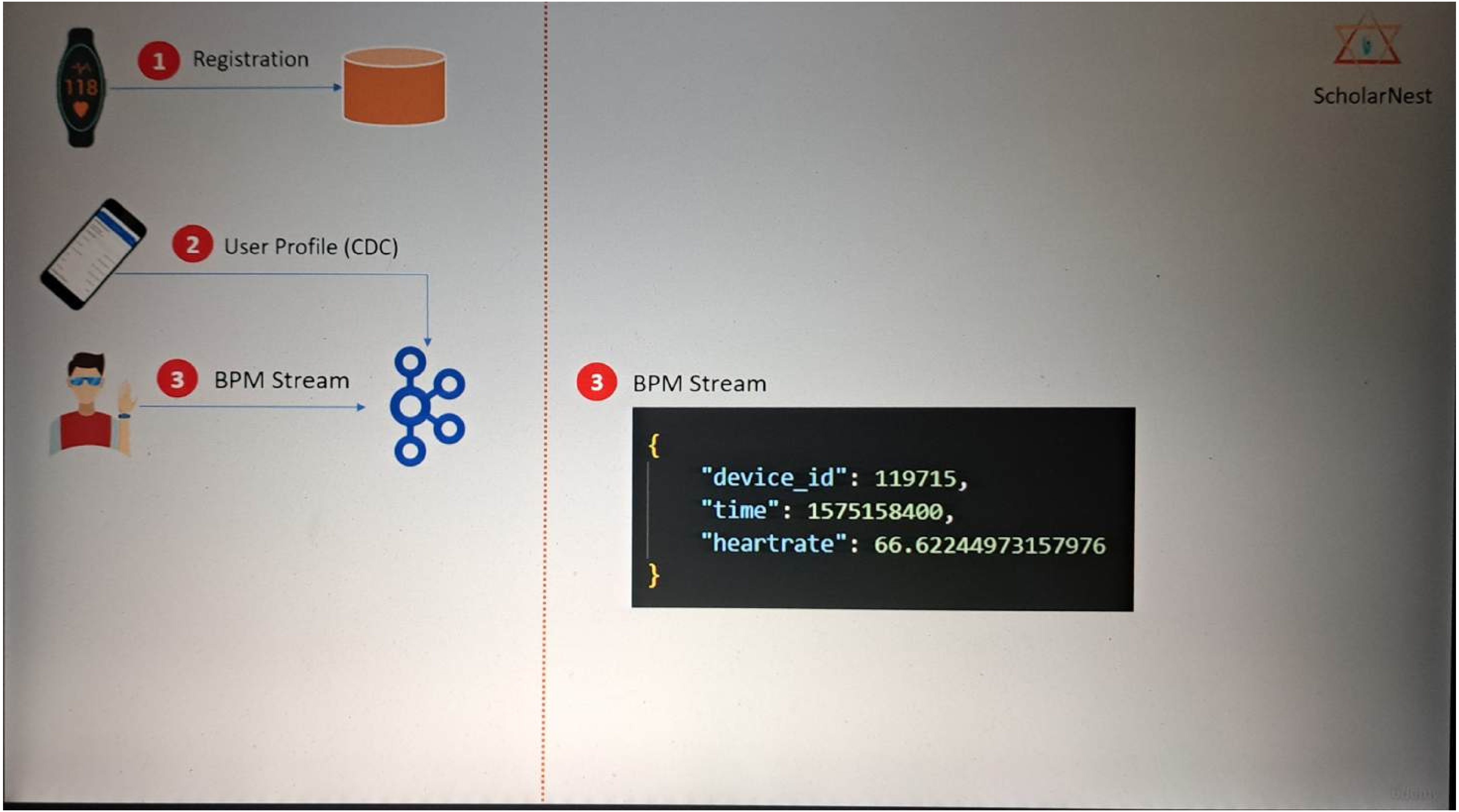


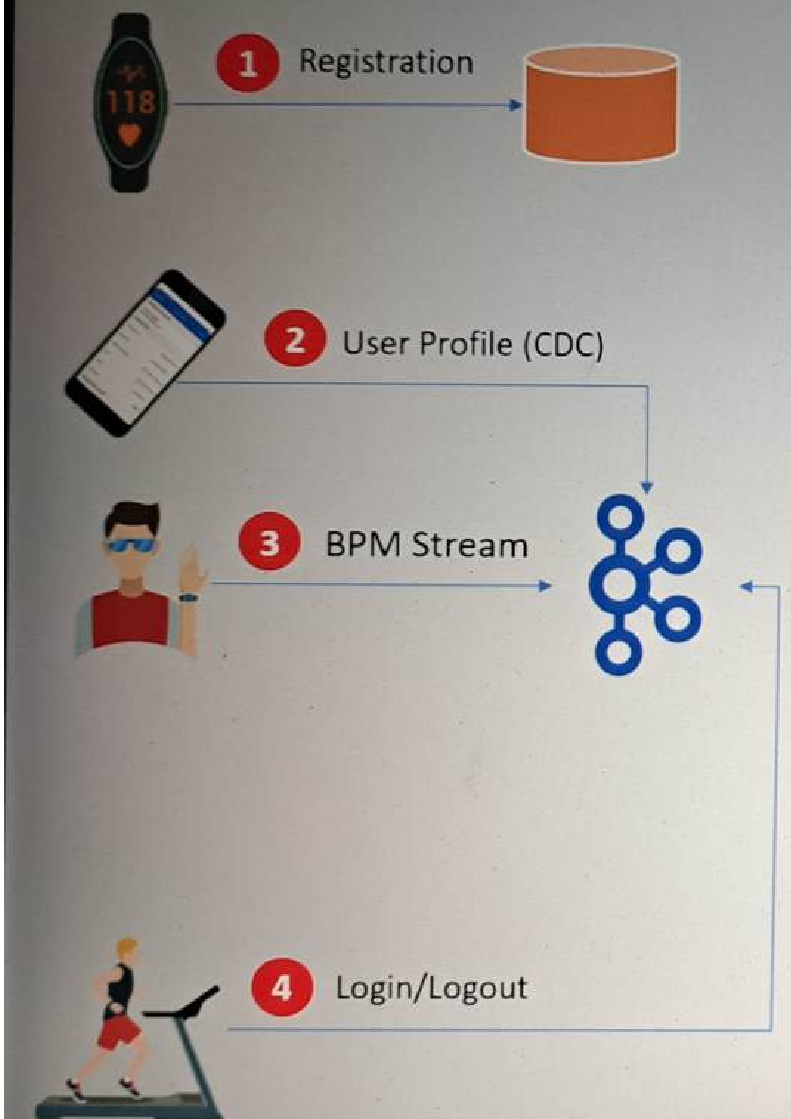

ScholarNest

1 Registration

| User ID | Device ID | Mac Address | Registration timestamp |
|---|---|---|---|
| 11745 | 190960 | 14:cd:d6:db:70:f6 | 1575763595.0 |
| 12140 | 141687 | ae:ec:f6:48:ca:f7 | 1558694443 |
| 12227 | 114131 | 57:24:ac:8c:75:ea | 1575816942.0 |
| 12474 | 118440 | 4c:c5:9f:cb:13:bd | 1575178082.0 |


1
Registration
118
2
User Profile (CDC)
New Profile
{
"user_id": 19548,
"update_type": "new",
"timestamp": 1576438349.0,
"dob": "02/12/1972",
"sex": "F",
"gender": "F",
"first_name": "Megan",
"last_name": "Daugherty",
"address": {
"street_address": "5641 Kelly Tunnel Apt. 584",
"city": "Los Angeles",
"state": "CA",
"zip": 90026
}
}
new
Profile Update
{
"user_id": 36164,
"update_type": "update",
"timestamp": 1576264662.0,
"dob": "10/08/1935",
"sex": "F",
"gender": "F",
"first_name": "Caitlin",
"last_name": "Miles",
"address": {
"street_address": "176 Sandra Lane Suite 564",
"city": "Carson",
"state": "CA",
"zip": 90746
}
}
update
ScholarNest
Delete
delete
{
"user_id": 42643,
"update_type": "delete",
"timestamp": 1575853777.0
}

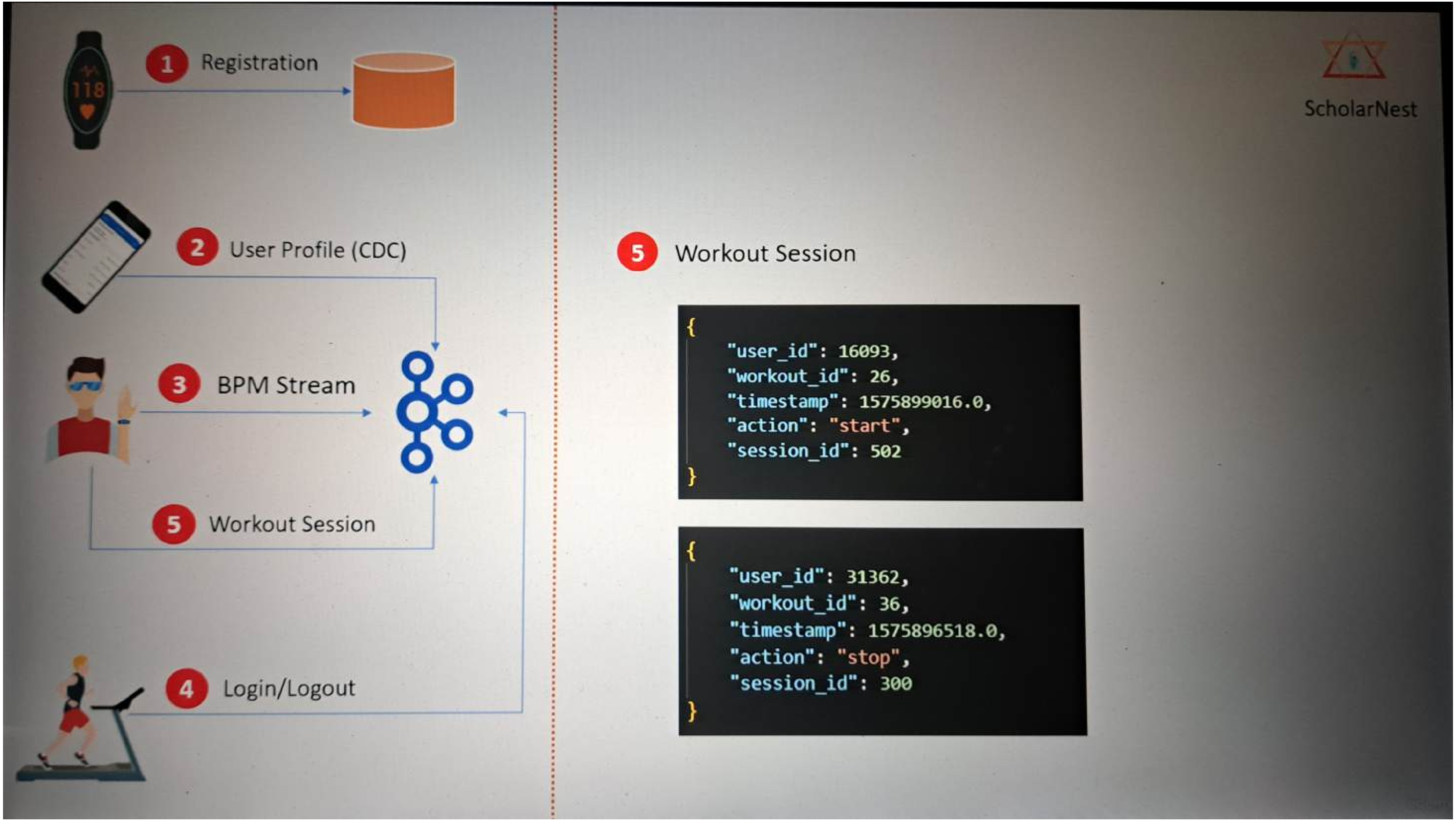

ScholarNest
1 Registration
118
2 User Profile (CDC)
3 BPM Stream
3 BPM Stream
{
"device_id": 119715,
"time": 1575158400,
"heartrate": 66.62244973157976
}

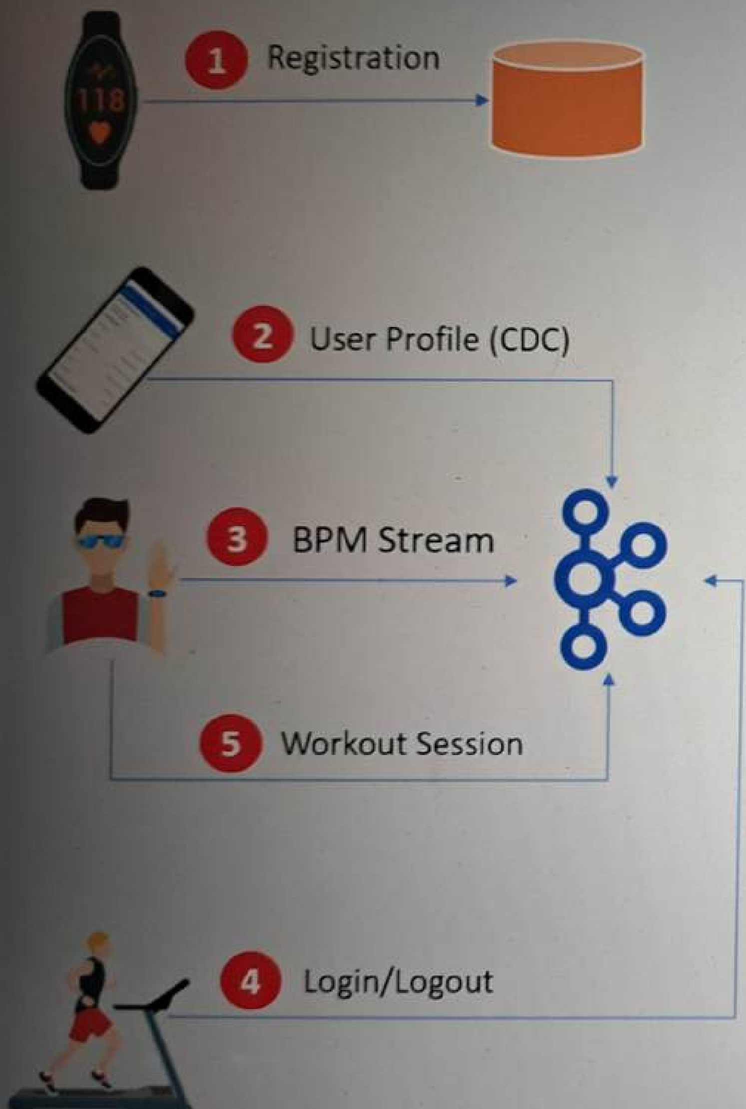



ScholarNest

4 Login/Logout

| Mac Address | Gym | Login | Logout |
|---|---|---|---|
| 14:cd:d6:db:70:f6 | 5 | 1575202519 | 1575208579 |
| ae:ec:f6:48:ca:f7 | 3 | 1575194342 | 1575200195 |
| 57:24:ac:8c:75:ea | 3 | 1575220992 | 1575225715 |
| 4c:c5:9f:cb:13:bd | 4 | 1575184296 | 1575191636 |


1 Registration
118
2 User Profile (CDC)
3 BPM Stream
5 Workout Session
4 Login/Logout
ScholarNest
5 Workout Session
{
"user_id": 16093,
"workout_id": 26,
"timestamp": 1575899016.0,
"action": "start",
"session_id": 502
}
{
"user_id": 31362,
"workout_id": 36,
"timestamp": 1575896518.0,
"action": "stop",
"session_id": 300
}




ScholarNest

- Design and implement a Lakehouse platform using a medallion architecture pattern.
- Collect and ingest data from the source system to your platform.
- Prepare the following analysis data sets for the data consumers
  - Workout BPM Summary
  - Gym Summary

ScholarNest

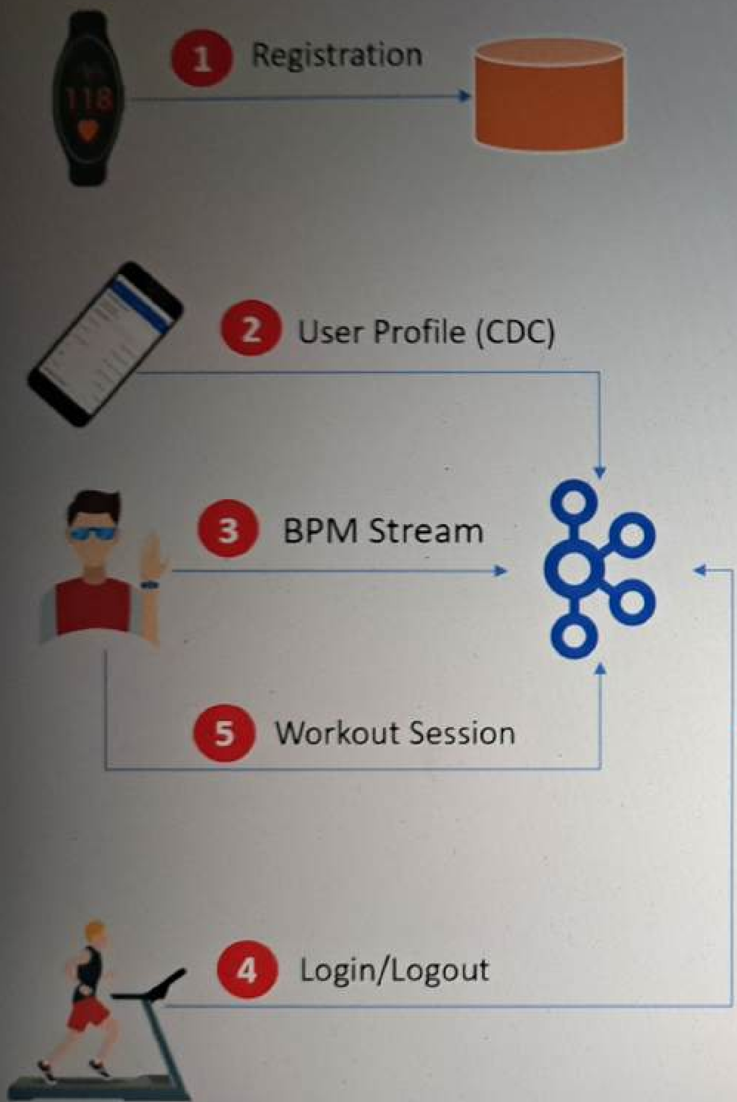
# Workout BPM Summary

| User id | Date | Workout id | Session id | Age | Sex | City | State | #Rec | Min bpm | Avg. bpm | Max bpm |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 12140 | 2019-12-11 | 5 | 478 | 18-25 | M | Pear blossom | CA | 205 | 86.97 | 114.89 | 176.84 |
| 12140 | 2019-12-11 | 6 | 479 | 18-25 | M | Pear blossom | CA | 206 | 77.76 | 112.73 | 181.24 |
| 12140 | 2019-12-12 | 9 | 480 | 18-25 | M | Pear blossom | CA | 294 | 78.66 | 120.67 | 165.53 |

# Gym Summary

| Gym | Mac Address | Date | Workouts | Minutes in Gym | Minutes Exercising |
|---|---|---|---|---|---|
| 4 | 32:60:3b:80:ed:8e | 2019-12-11 | [1, 18] | 107.23 | 96.00 |
| 4 | 32:60:3b:80:ed:8e | 2019-12-12 | [49, 32] | 108.60 | 100.26 |
| 4 | 32:60:3b:80:ed:8e | 2019-12-15 | [20, 17] | 92.63 | 78.93 |

ScholarNest




1. Design a secure Lakehouse platform with dev, test, and production environments
2. Decouple data ingestion from data processing
3. Support batch and streaming workflows
4. Automate integration testing
5. Automated deployment pipeline for test and production environments